SuRe: Summarizing Retrievals using Answer Candidates for Open-domain QA of LLMs

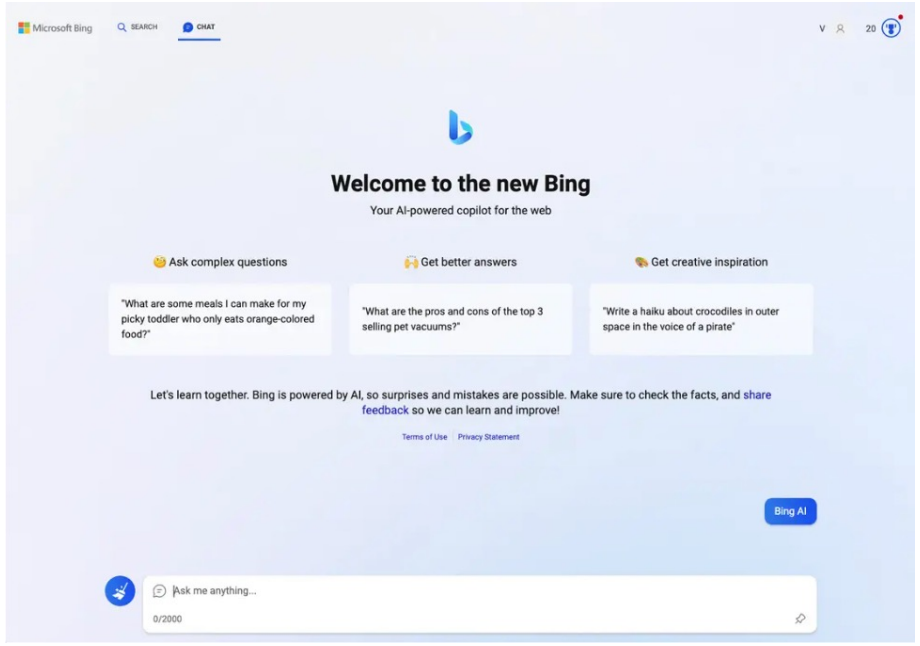
Jaehyung Kim[1], Jaehyun Nam[2], Sangwoo Mo[3], Jongjin Park[2],
Sang-Woo Lee[4], Minjoon Seo[2], Jung-Woo Ha[4], and Jinwoo Shin[2]

ICLR 2024

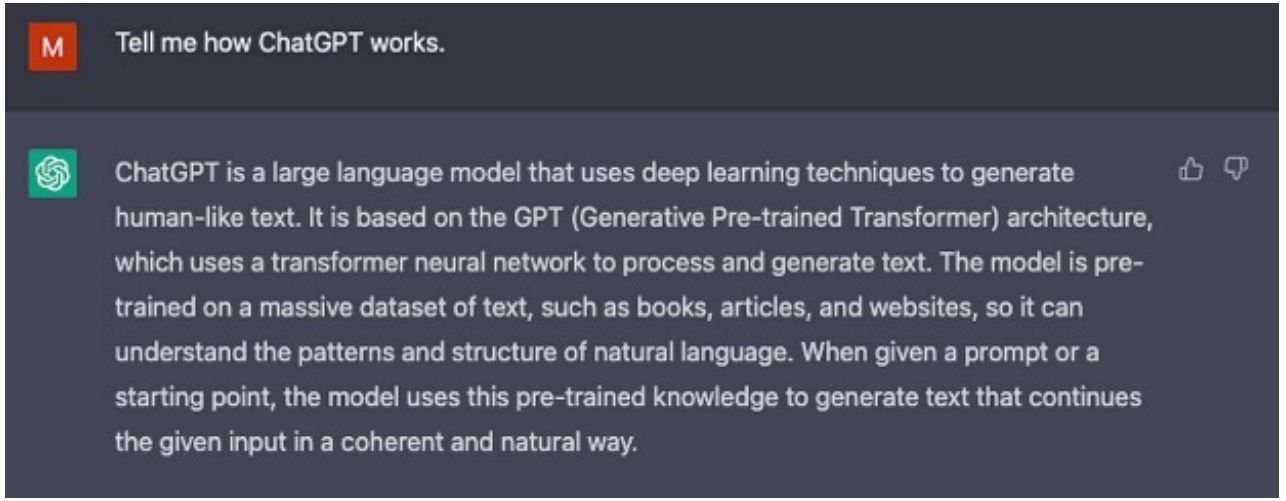
[1] CMU, [2] KAIST, [3] University of Michigan, [4] Naver

Success of Large Language Models (LLMs)

- LLMs become **core technology** in many real-world application



Search engine (Bing AI)[1]

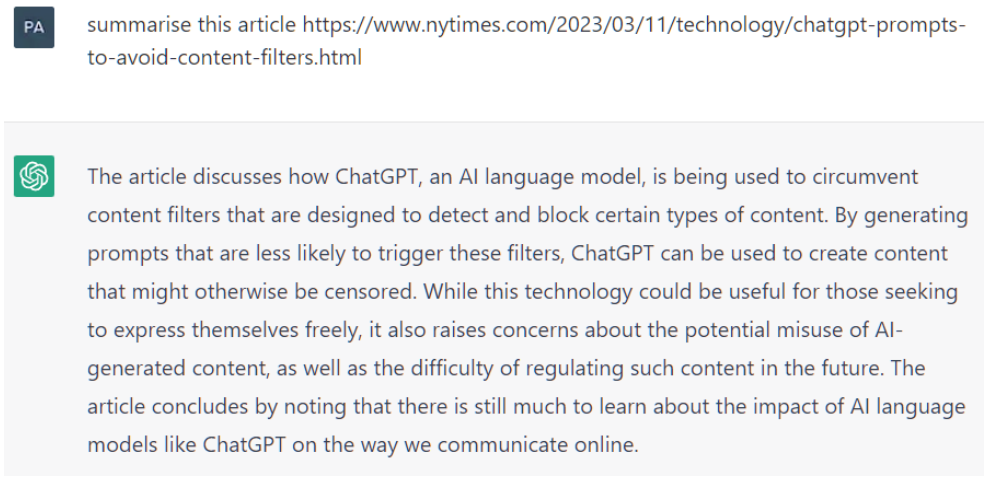


Chatbot (ChatGPT)[2]

[1] Quy et al., Evaluation of Chatgpt and Microsoft Bing ai Chat Performances on Physics Exams of Vietnamese National High School Graduation Examination., arXiv:2306
[2] OpenAI., https://openai.com/blog/chatgpt., 2022

Challenges of Large Language Models

- Despite its great success, LLMs still suffer from **critical limitations**
  - Non-factual but seemingly plausible generation, *i.e.*, hallucinations[1]
  - Difficulty in integrating up-to-date knowledge



Hallucination of LLMs[2]



Struggle to incorporate up-to-date knowledge

[1] Welleck et al., Neural Text Generation with Unlikelihood Training., ICLR 2020
[2] https://en.wikipedia.org/wiki/Hallucination_%28artificial_intelligence%29

Retrieval Augmentation

- **Retrieving** and incorporating relevant knowledge → promising solution
  - *Retrieval* is effective way to improve QA system, *i.e.*, open-domain QA (ODQA)[1,2]

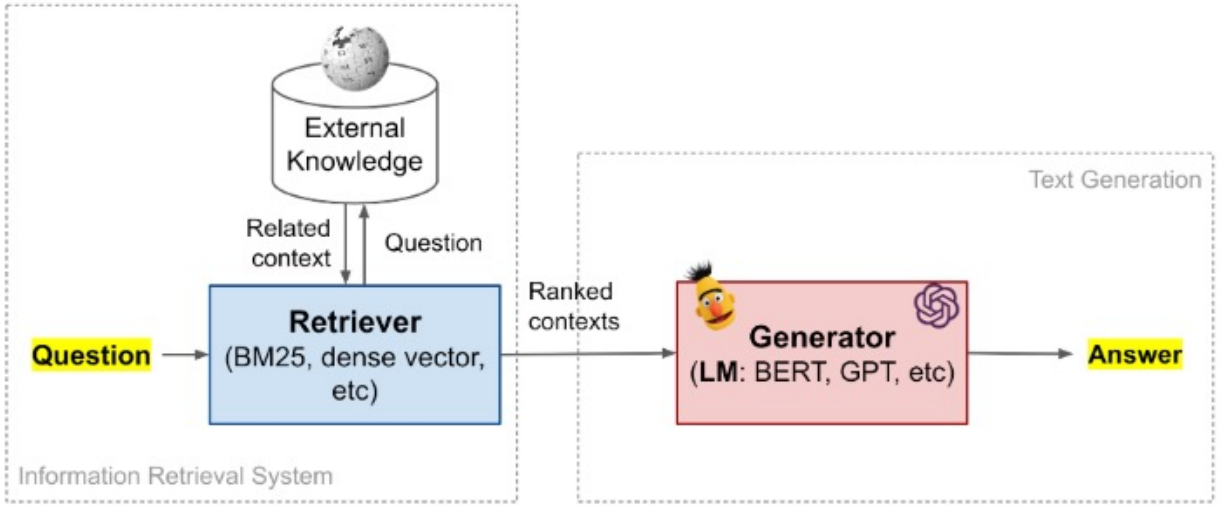
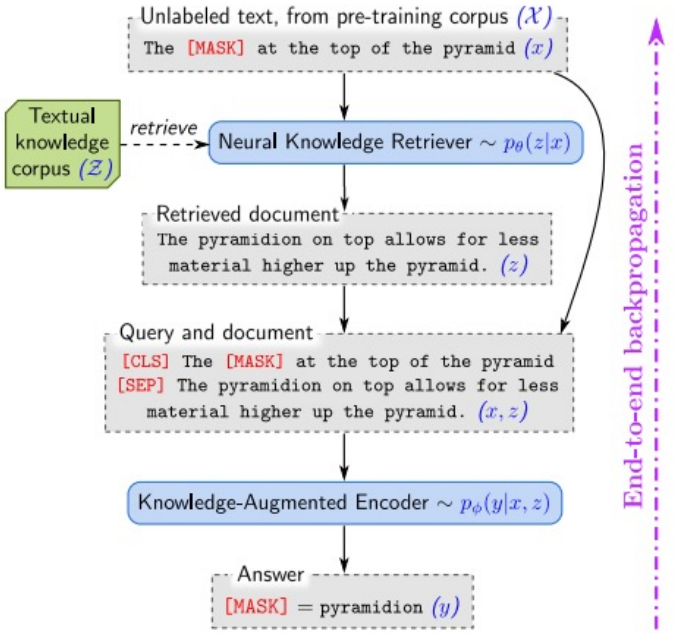


Illustration of retrieve-and-read system for ODQA[1]

[1] https://lilianweng.github.io/posts/2020-10-29-odqa/
[2] Karpukhin et al., Dense Passage Retrieval for Open-Domain Question Answering., EMNLP 2020

Retrieval Augmentation

- **Retrieving** and incorporating relevant knowledge → promising solution
  - *Retrieval* is effective way to improve QA system, *i.e.*, open-domain QA (ODQA)[1,2]
  - *Retrieval-augmented LLMs* outperforms ordinary LLMs[3,4]



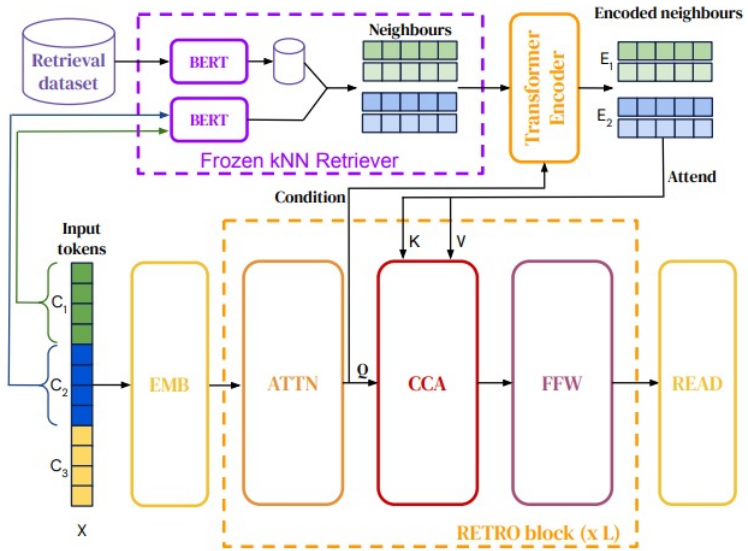
Name	NQ (79k/4k)	WQ (3k/2k)	CT (1k /1k)	# params
BERT-Baseline (Lee et al., 2019)	26.5	17.7	21.3	110m
T5 (base) (Roberts et al., 2020)	27.0	29.1	-	223m
T5 (large) (Roberts et al., 2020)	29.8	32.2	-	738m
T5 (11b) (Roberts et al., 2020)	34.5	37.4	-	11318m
DrQA (Chen et al., 2017)	-	20.7	25.7	34m
HardEM (Min et al., 2019a)	28.1	-	-	110m
GraphRetriever (Min et al., 2019b)	31.8	31.6	-	110m
PathRetriever (Asai et al., 2019)	32.6	-	-	110m
ORQA (Lee et al., 2019)	33.3	36.4	30.1	330m
Ours ($\mathcal{X}$ = Wikipedia, $\mathcal{Z}$ = Wikipedia)	39.2	40.2	**46.8**	330m
Ours ($\mathcal{X}$ = CC-News, $\mathcal{Z}$ = Wikipedia)	**40.4**	**40.7**	42.9	330m

Illustration of REtrieval-Augmented Language Model (REALM)[3]

[1] https://lilianweng.github.io/posts/2020-10-29-odqa/
[2] Karpukhin et al., Dense Passage Retrieval for Open-Domain Question Answering., EMNLP 2020
[3] Guu et al., REALM: Retrieval-Augmented Language Model Pre-Training., ICML 2020
[4] Borgeaud et a., Improving Language Models by Retrieving from Trillions of Tokens., ICML 2022

Challenges with Retrieval Augmented LLMs

- Retrieval-augmented LLMs are typically constructed with **finetuning**[1,2]
  - But, recent large scale & black-box nature make it less attractive & infeasible

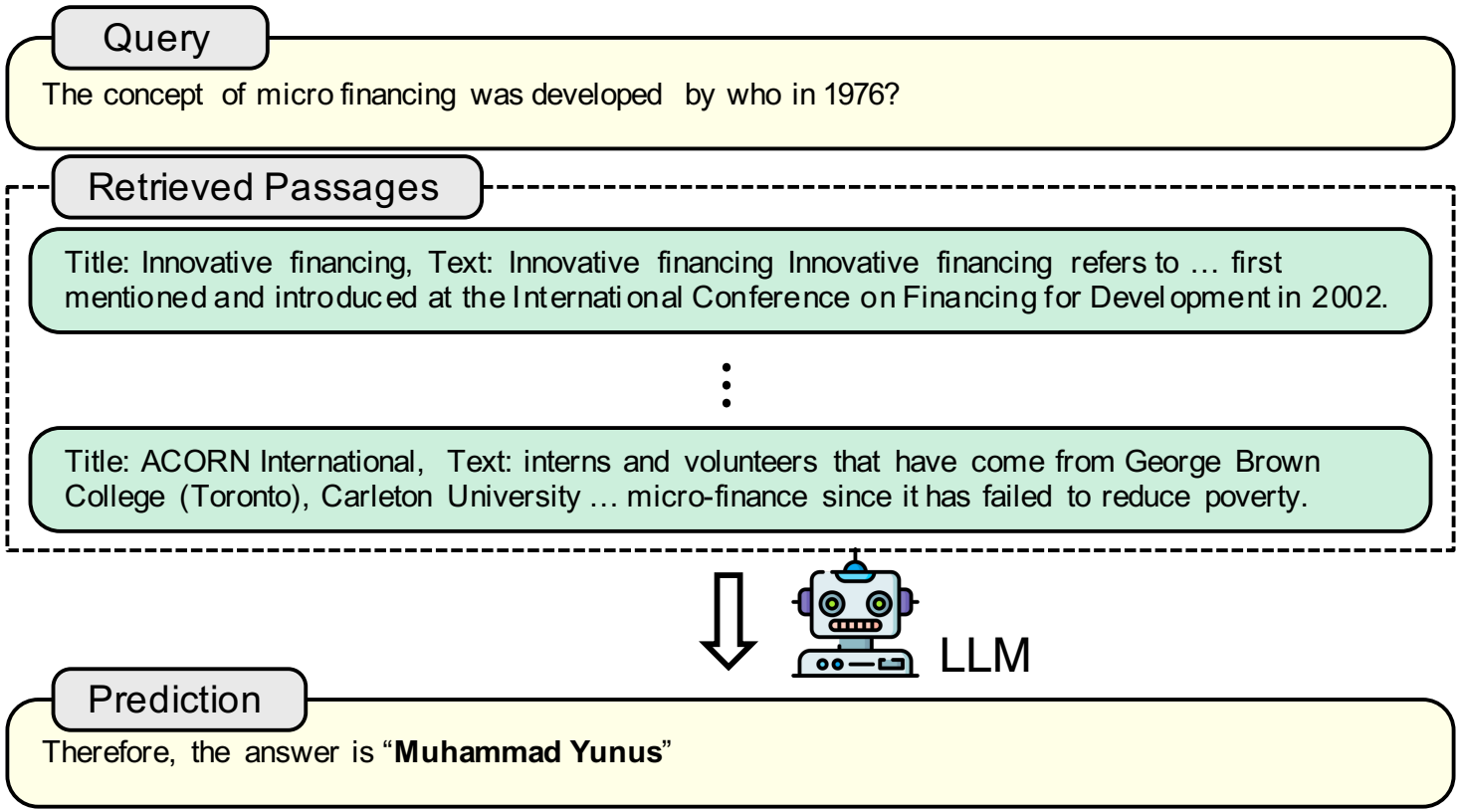


Modification of architecture and training to learn retrieval augmentation[2]

[1] Guu et al., REALM: Retrieval-Augmented Language Model Pre-Training., ICML 2020
[2] Borgeaud et al., Improving Language Models by Retrieving from Trillions of Tokens., ICML 2022

Challenges with Retrieval Augmented LLMs

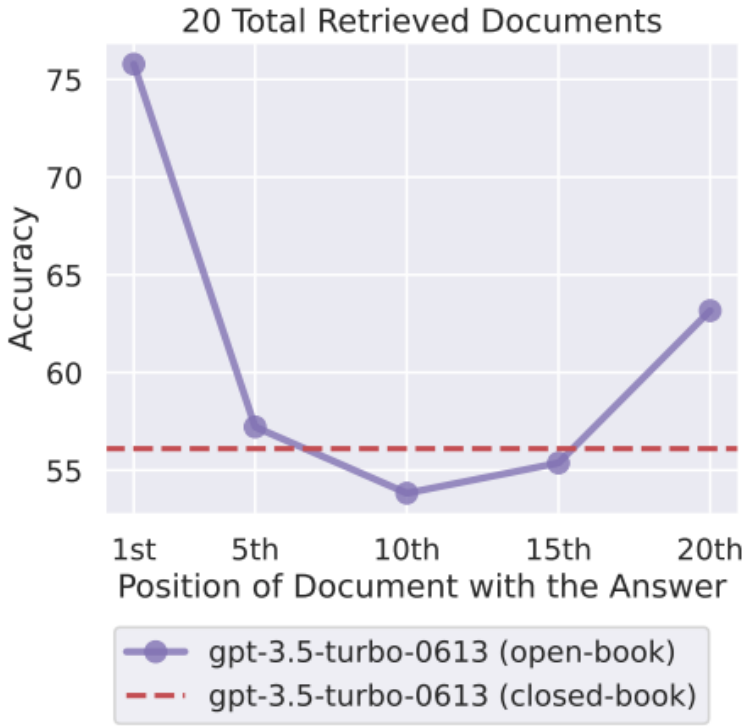
- **Prompting** is simple and efficient way to augment retrieved passages[1]



[1] Shi et al., REPLUG: Retrieval-Augmented Black-Box Language Models., arXiv:2301
[2] Liu et al., Lost in the Middle: How Language Models Use Long Contexts., arXiv:2307

Challenges with Retrieval Augmented LLMs

- **Prompting** is simple and efficient way to augment retrieved passages[1]
  - Naïve approach (*e.g.*, appending) could be limited in fully exploiting the retrieval[2]



LLMs struggle to handle long retrieved passages[2]

[1] Shi et al., REPLUG: Retrieval-Augmented Black-Box Language Models., arXiv:2301
[2] Liu et al., Lost in the Middle: How Language Models Use Long Contexts., arXiv:2307

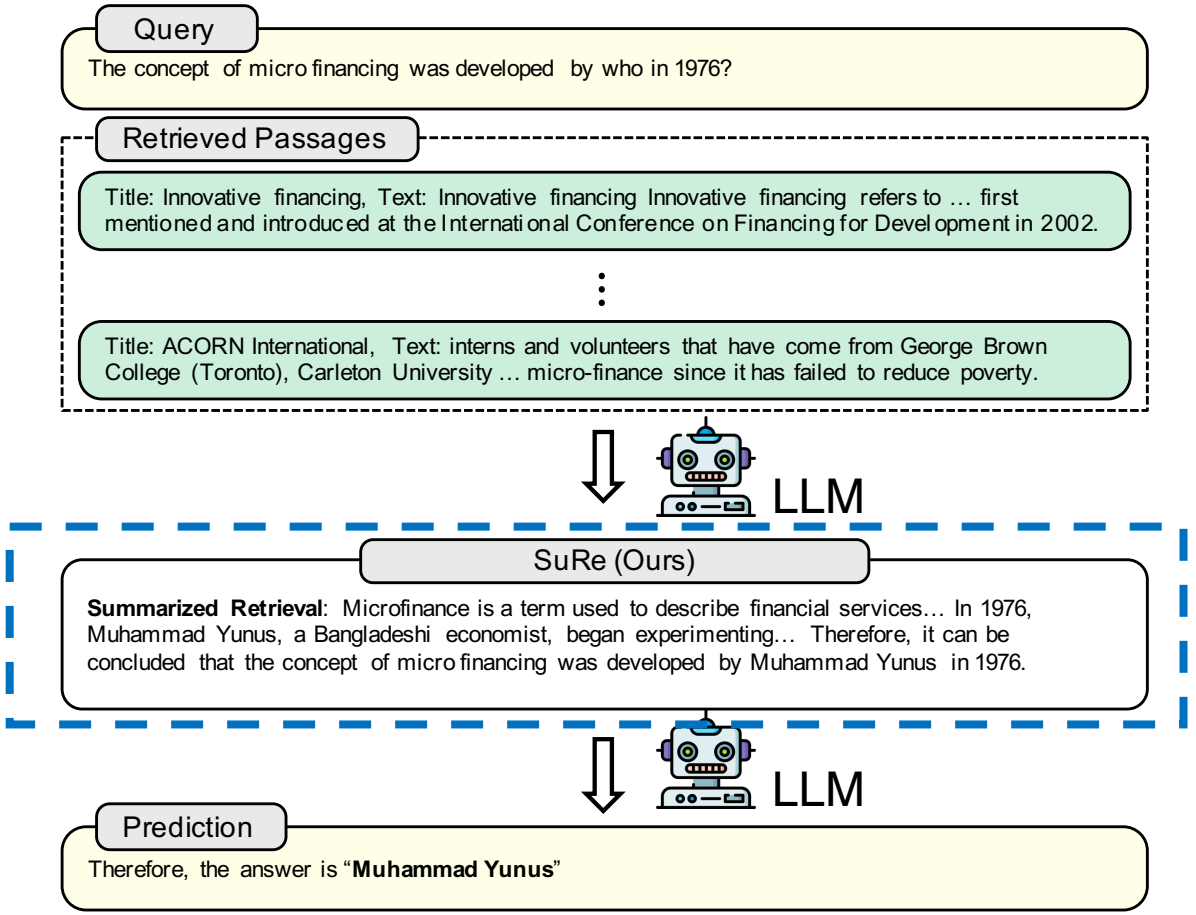
Research Goal

- **Prompting** is simple and efficient way to augment retrieved passages[1]
  - Naïve approach (*e.g.*, appending) could be limited in fully exploiting the retrieval[2]

Research Goal: a simple yet effective framework based on prompting to improve ODQA with LLMs

Proposed Method: SuRe

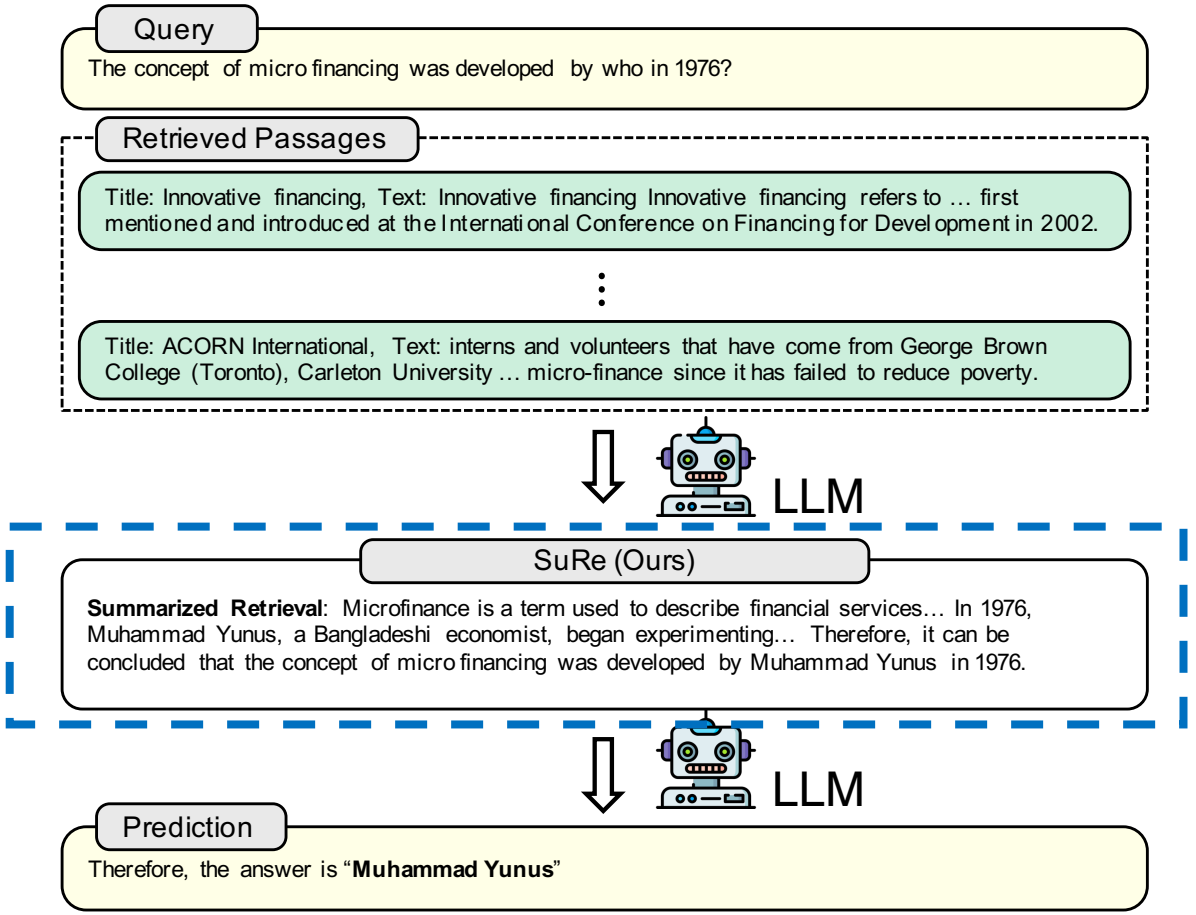
- **SuRe**: Summarizing Retrievals using Answer Candidates



Example of Retrieval Augmentation of LLMs via Proposed SuRe Framework

Proposed Method: SuRe

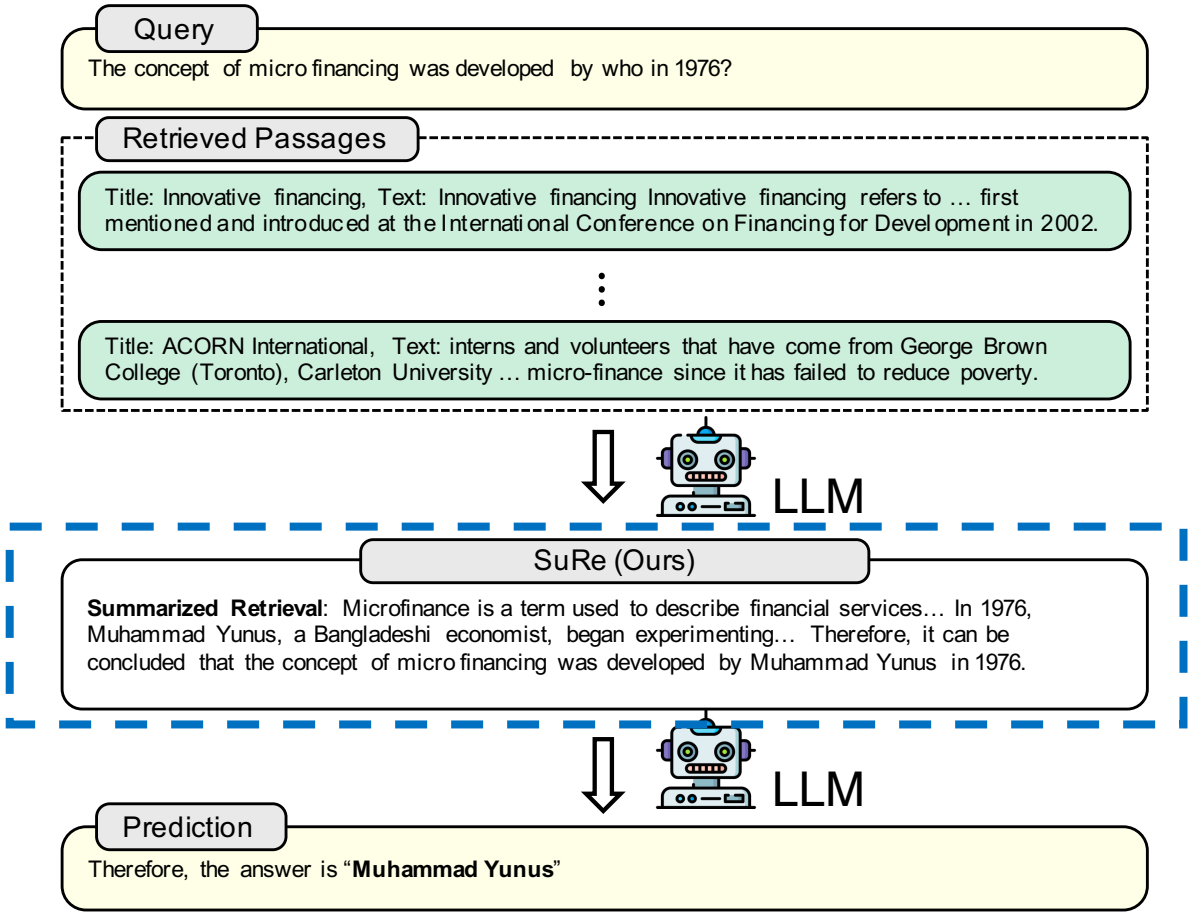
- **SuRe**: Summarizing Retrievals using Answer Candidates
  - Key idea. Select answer that is well-supported by summarization of retrievals (explicit rationale)



Example of Retrieval Augmentation of LLMs via Proposed SuRe Framework

Proposed Method: SuRe

- **SuRe**: Summarizing Retrievals using Answer Candidates
  - Key idea. Select answer that is well-supported by summarization of retrievals



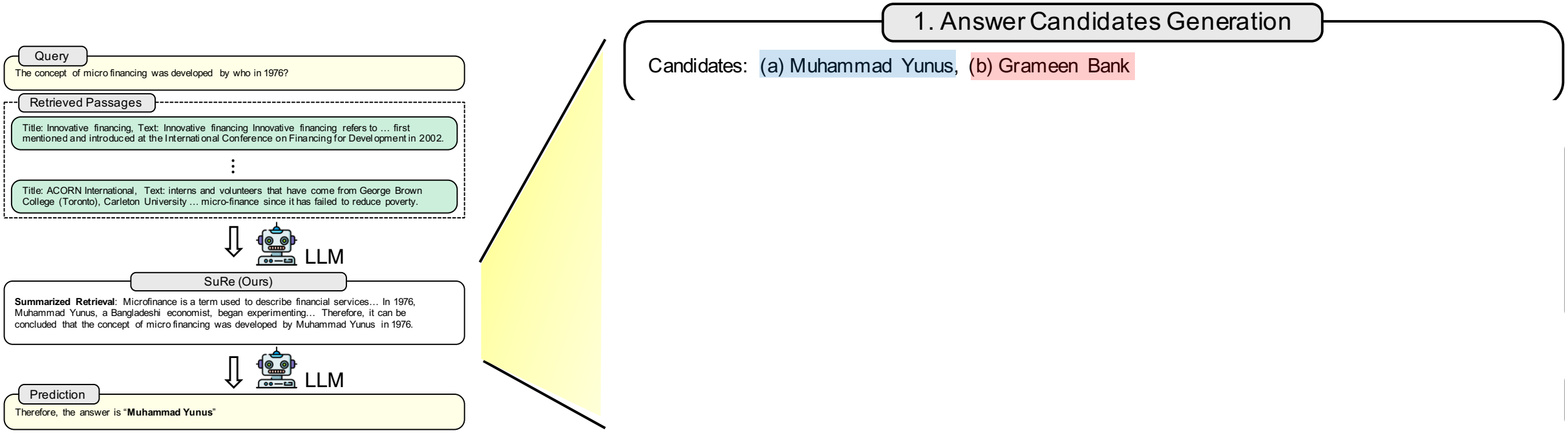
Example of Retrieval Augmentation of LLMs via Proposed SuRe Framework

Proposed Method: SuRe

- SuRe conducts **three steps** sequentially (via zero-shot prompting)

Proposed Method: SuRe

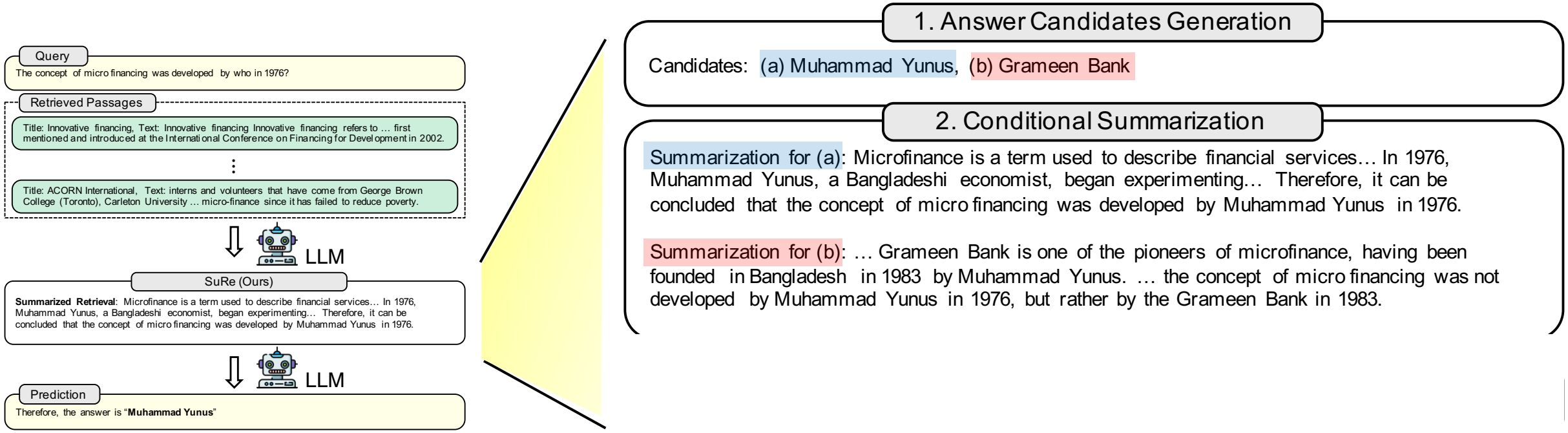
- SuRe conducts **three steps** sequentially (via zero-shot prompting)
  1. Answer candidates generation



Overall illustration of SuRe Framework

Proposed Method: SuRe

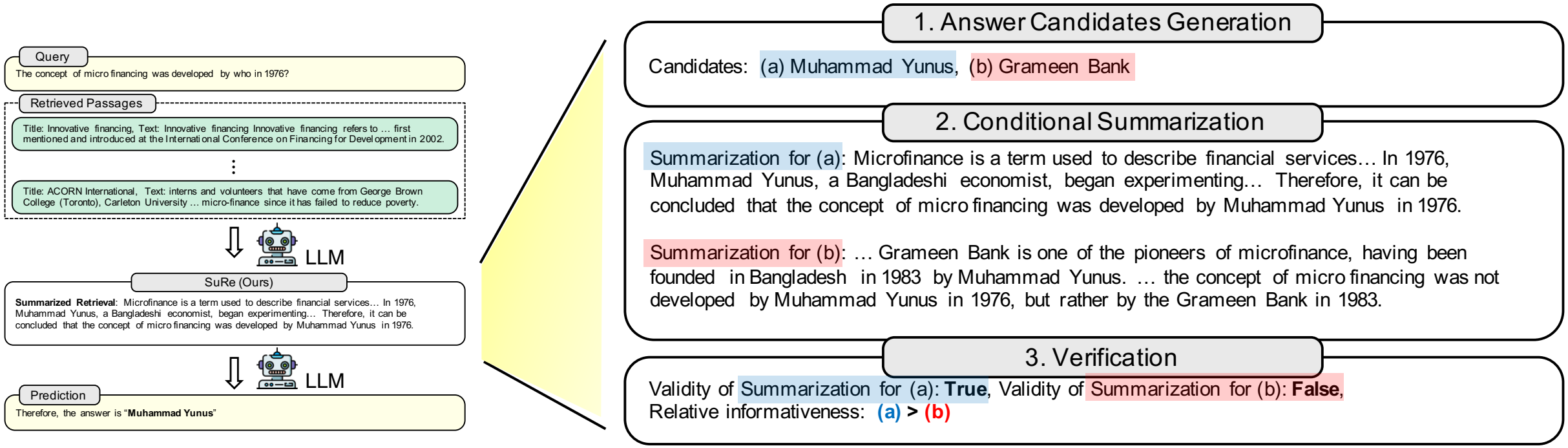
- SuRe conducts **three steps** sequentially (via zero-shot prompting)
  1. Answer candidates generation
  2. Conditional summarization



Overall illustration of SuRe Framework

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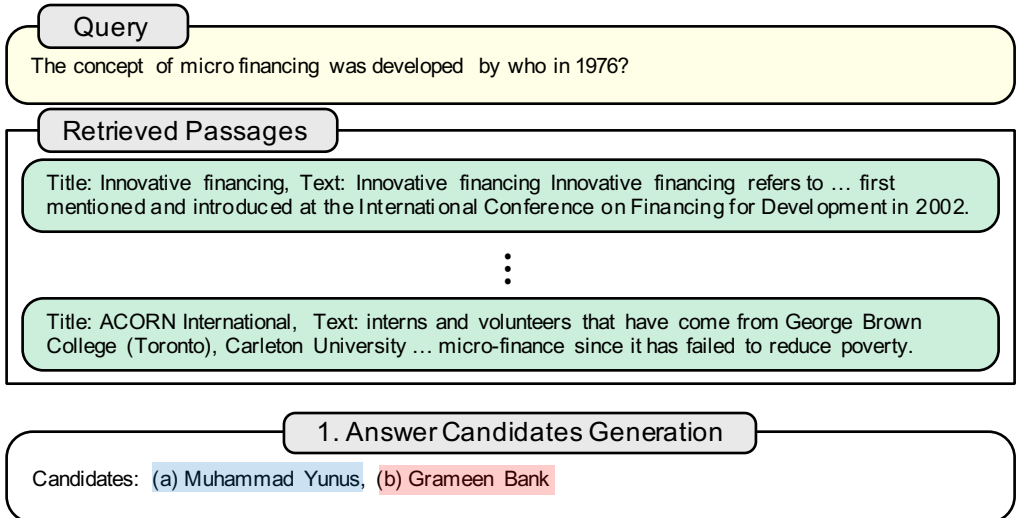
- SuRe conducts **three steps** sequentially (via zero-shot prompting)
  1. Answer candidates generation
  2. Conditional summarization
  3. Selection via verification (validity & ranking)



Overall illustration of SuRe Framework

SuRe: Answer Candidates Generation

- SuRe first generate K answer candidates from question and retrieval
  - Compare to random sampling, we observe *direct prompting is more effective*
  - We use fixed K=2 during experiments, as K > 2 is quite inefficient



Listing 1 Prompt for answer candidates generation.

```
f'''
Below are N passages related to the question at the end. After reading
↪  the passages, provide two correct candidates for the answer to the
↪  question at the end. Each answer should be in the form: (a) xx, (b)
↪  yy, and should not exceed 3 words for each candidate.

Passage #1 Title: {Passage #1 Title}
Passage #1 Text: {Passage #1 Text}

...

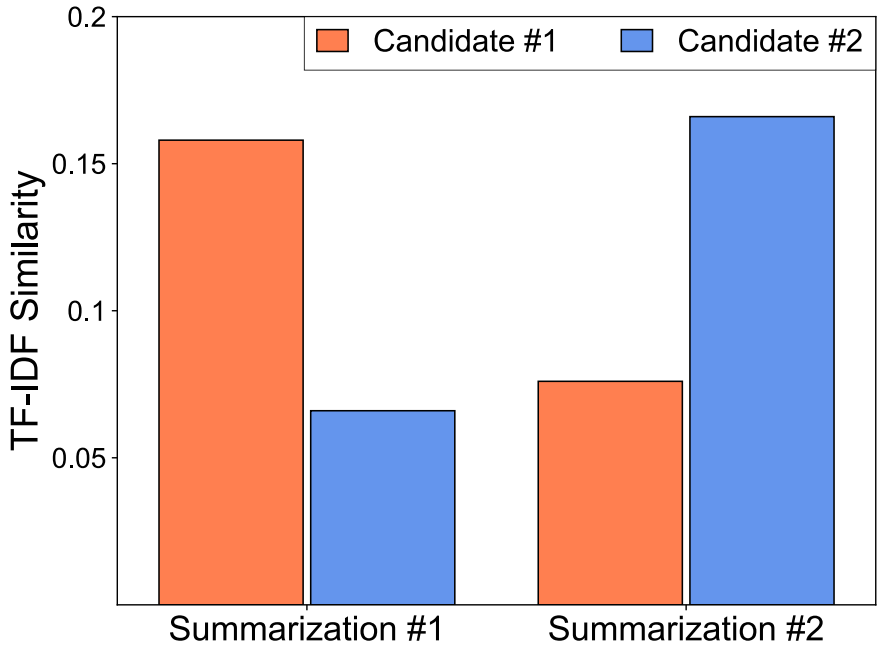
Passage #N Title: {Passage #N Title}
Passage #N Text: {Passage #N Text}

Question: {Question}

Answer:
'''
```

SuRe: Conditional Summarization

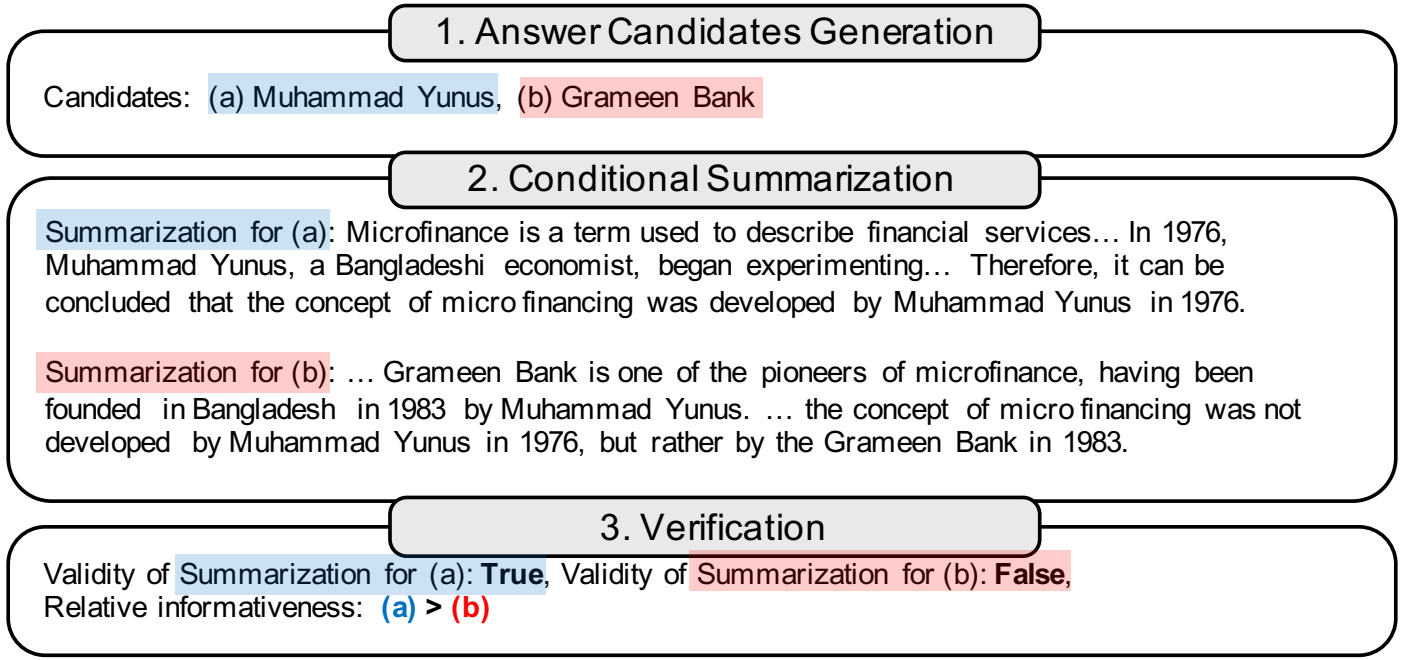
- SuRe then summarize retrieval conditioned on each candidate
  - *i.e.*, question, retrieved passages, and answer candidate → summarization
  - Conditional summarization includes *specific contexts* supporting given candidate



TF-IDF similarity between candidates and conditional summarizations

SuRe: Selection via Verification

- Lastly, SuRe verify summarizations and select most plausible candidate



SuRe: Selection via Verification

- Lastly, SuRe verify summarizations and select most plausible candidate
  - Two scores: (1) Validity of summarization

Listing 3 Prompt for instance-wise validation.

```
f'''
Question: {Question}

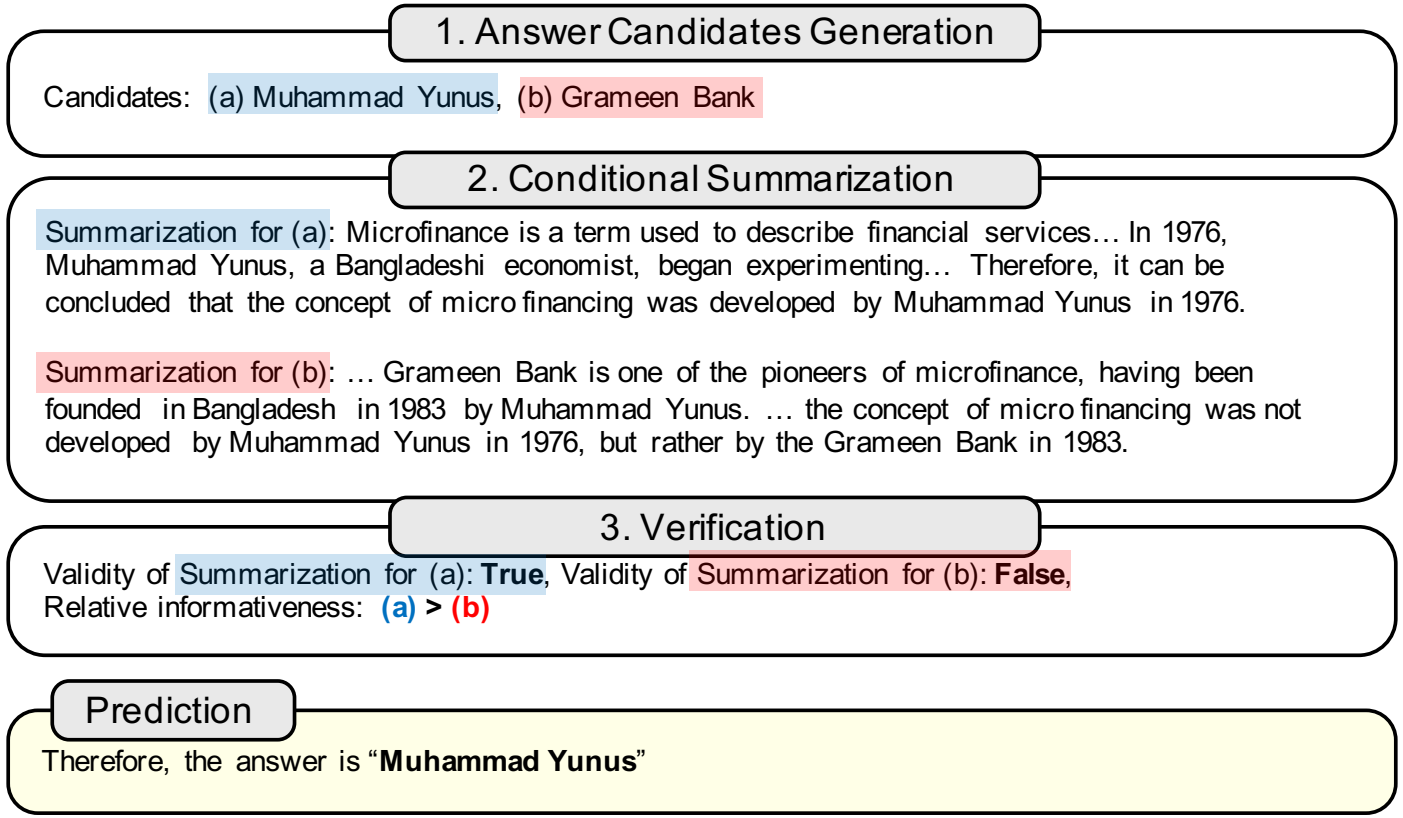
Prediction: {Prediction}

Passage: {Passage}

Does the passage correctly support the prediction? Choices: [True,
↪  False]. Answer:
'''
```

SuRe: Selection via Verification

- Lastly, SuRe verify summarizations and select most plausible candidate
  - Two scores: (1) Validity of summarization & (2) Ranking between summarizations
  - Candidate with highest scored summarization → **Answer**



Experiments: Comparison with Baselines

- **ChatGPT** with 10 retrieved passages from **BM25**
  - Augmenting retrieved passages with prompting (*Base*) is effective
  - Other baselines are ineffective due to challenging setup (zero-shot, black-box API)
  - SuRe significantly improve *Base* with a large margin (4.6% in EM & 4.0% in F1)

Methods / Datasets	NQ	WebQ	2Wiki	HotpotQA	Average
No retrieval	27.6 / 39.0	25.0 / **38.8**	21.4 / 24.8	22.2 / 31.9	24.1 / 33.6
Base	28.4 / 38.8	19.6 / 32.5	27.4 / 32.8	30.8 / 40.3	26.6 / 36.1
Rerank	24.8 / 33.9	18.8 / 30.6	23.0 / 28.4	27.8 / 37.4	23.6 / 32.6
RePlug	26.0 / 35.3	18.8 / 31.5	23.6 / 28.5	28.0 / 37.9	24.1 / 33.3
Selection-inference	24.3 / 32.8	17.3 / 28.6	22.6 / 29.5	30.8 / 39.6	23.8 / 32.6
Chain-of-thoughts	22.3 / 31.4	15.2 / 27.8	19.6 / 22.5	25.6 / 31.8	20.7 / 28.4
Self-verification	25.2 / 35.4	16.1 / 28.5	23.2 / 30.5	31.6 / 41.8	24.0 / 34.1
SuRe (Ours)	**33.5 / 42.3**	**25.1** / 36.6	**32.8 / 38.1**	**33.2 / 43.4**	**31.2 / 40.1**

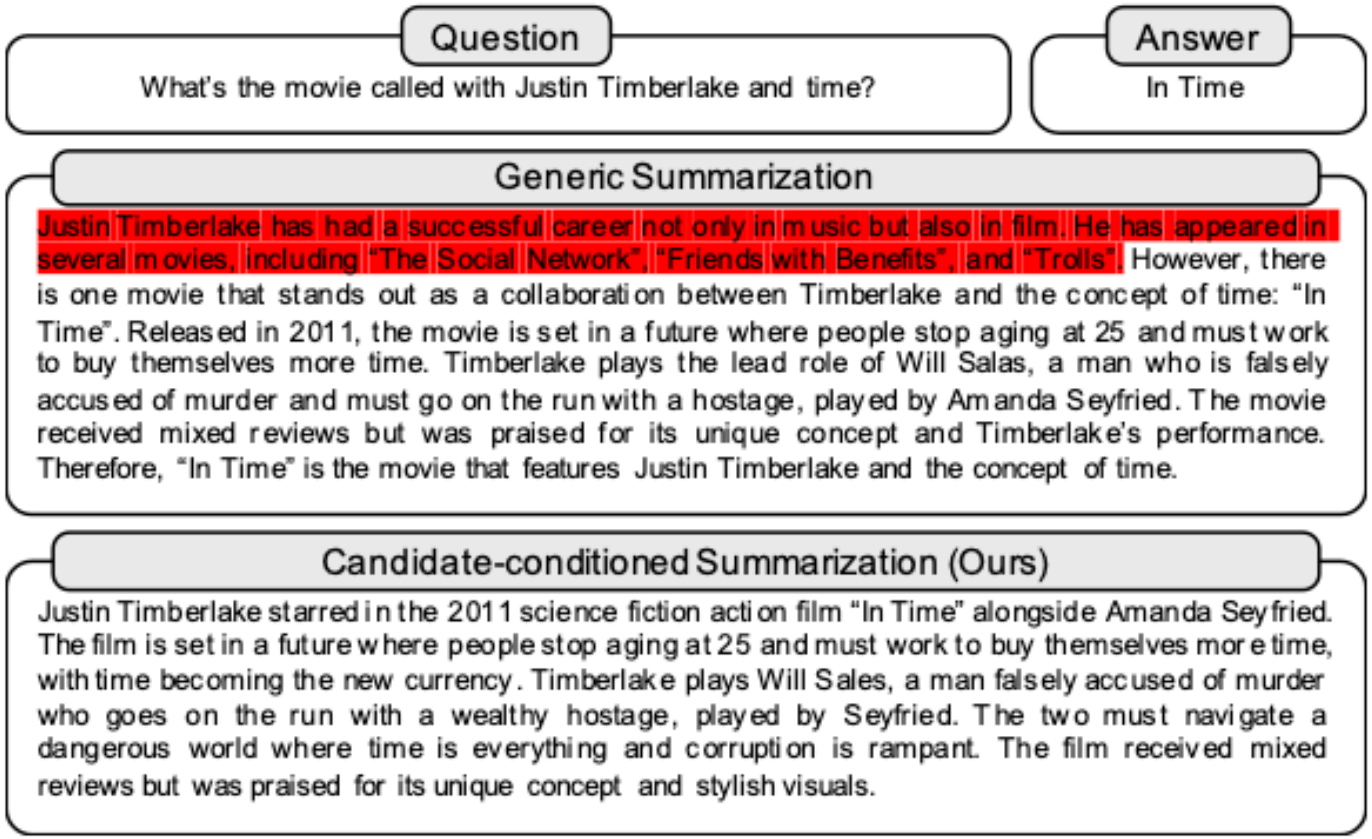
Experiments: Compatibility of SuRe

- Evaluation on **different LLMs and retrieval methods**
  - Naïve prompting (Base) is compared as baseline
  - SuRe is consistently effective for all setups with 4.6% EM improvement in average
  - SuRe successfully improves LLaMA2-chat → Usefulness for open sourced LLMs!

	ChatGPT						GPT-4		LLaMA2-chat	
Datasets	BM25	+ SuRe	DPR	+ SuRe	Contriever	+ SuRe	BM25	+ SuRe	BM25	+ SuRe
NQ	28.4	**33.5**	36.1	**41.0**	35.8	**40.4**	30.2	**32.4**	18.6*	**30.4***
WebQ	19.6	**25.1**	23.2	**27.3**	22.5	**28.7**	21.5	**21.7**	16.0*	**24.0***
2Wiki	27.4	**32.8**	19.2	**21.4**	27.2	**32.6**	34.8	**38.2**	20.2	**27.8**
HotpotQA	30.8	**33.2**	25.6	**27.4**	32.2	**33.6**	34.8	**40.6**	24.0	**28.0**
Average	26.6	**31.2**	26.0	**29.3**	29.4	**33.8**	30.3	**33.2**	19.7	**27.6**

Experiments: Additional Advantages of SuRe

- Not only answer, but SuRe also generates **conditional summarization**
  - Then, is this conditional summarization really a *good rationale for answer*?



Qualitative comparison between SuRe's summarization and generic (i.e., unconditional) summarization

Experiments: GPT-4 & Human Evaluations

- We ask GPT-4 to select more informative & plausible sum. as rationale[1]

```
f'''
Question: Given the following summaries for the target question,
↪  determine which one is more informative and plausible as rationale
↪  to support a given target question-answer pair.

Summary 1: {Summary 1}

Summary 2: {Summary 2}

Target Question: {Question}

Target Answer: {Answer}

Your Task:
Identify which summary (Summary 1 or Summary 2) is more informative and
↪  plausible as rationale to support a given answer at hand. Choices:
↪  [Summary 1, Summary 2].

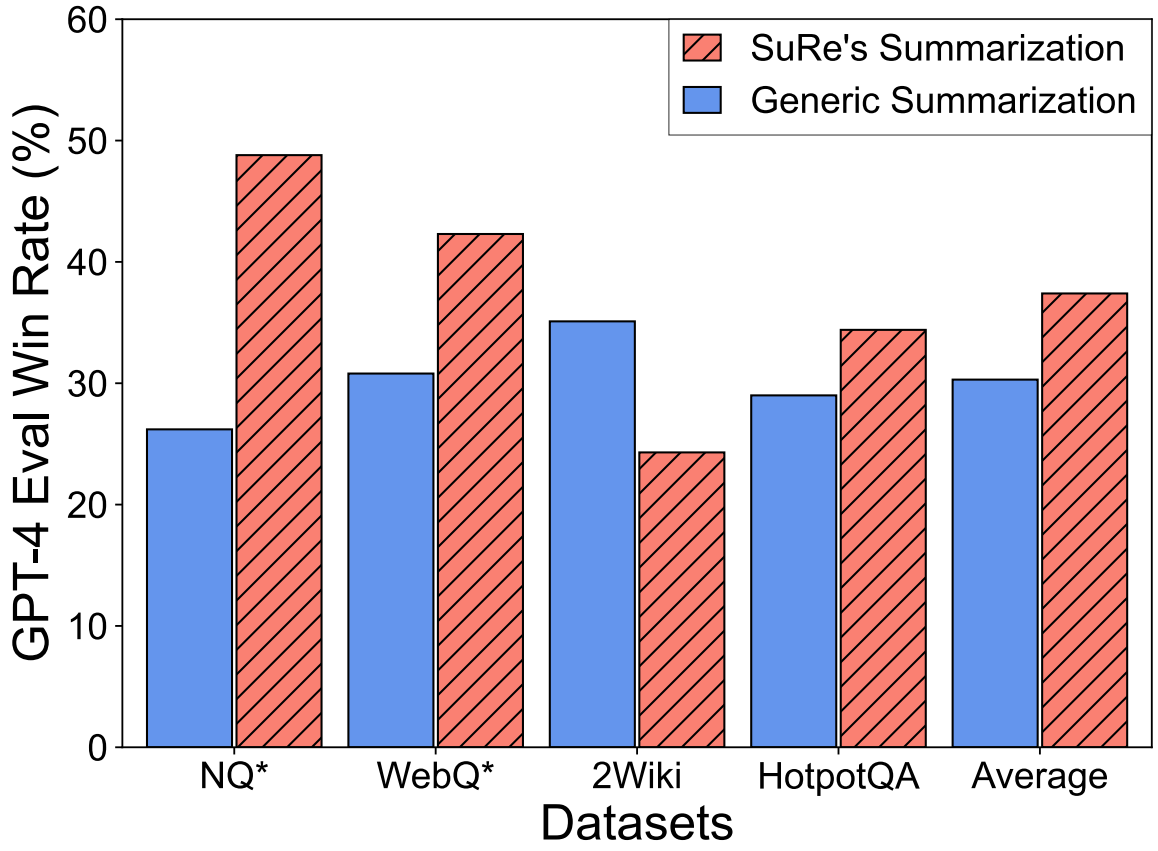
Answer:
'''
```

Designed Prompt for GPT-4 Evaluation between Summarizations

[1] Liu et al., G-Eval: NLG Evaluation using GPT-4 with Better Human Alignment., arXiv:2303

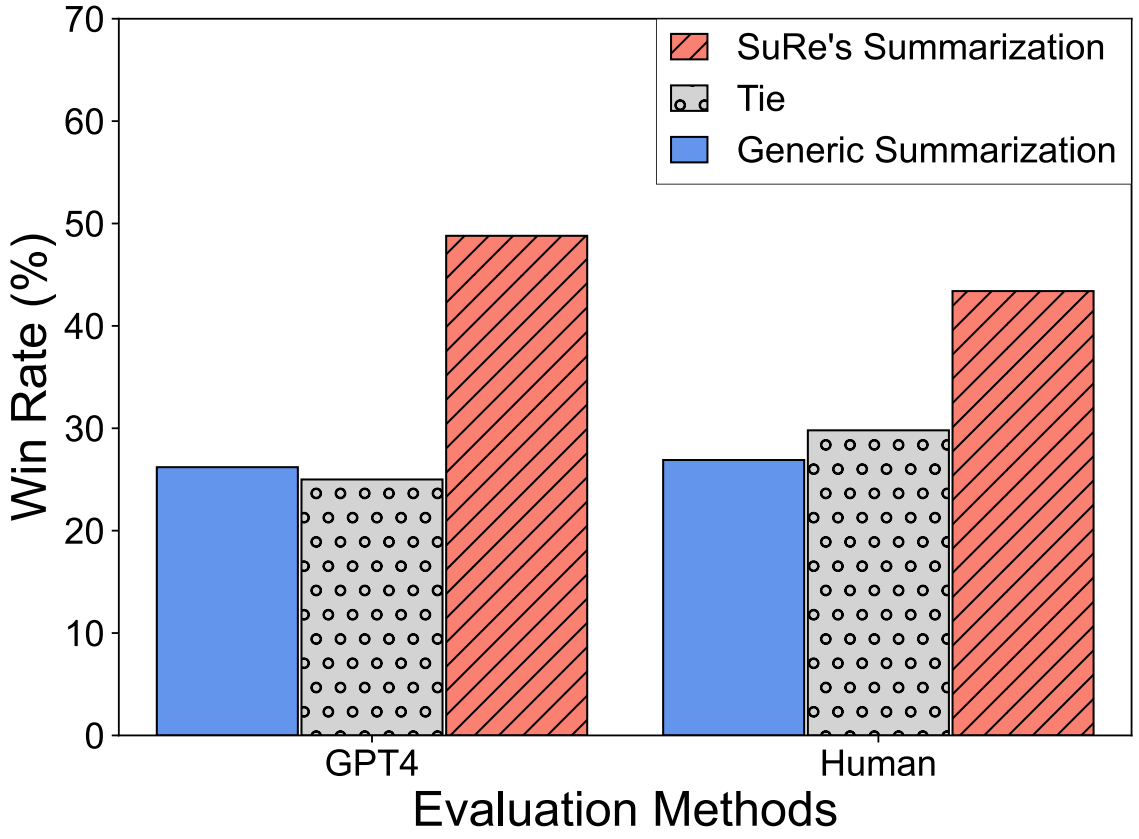
Experiments: GPT-4 Evaluation

- **GPT-4 prefers SuRe's** summarization than generic summarization
  - Setup. We only use questions that both summarizations correctly predict
  - Result. Average: **37.4%** (SuRe) vs 30.3 % (Generic)



Experiments: Human Evaluation

- **Human also prefers SuRe's** summarization than generic one
  - Setup. 84 NQ questions that both summarizations correctly predict (7 experts)
  - Result. **43.4%** (SuRe) vs 26.9 % (Generic)



Thank you for attention 🙏